

Searching Algorithms Programs in C

1.Program to Implement Binary Search Using C

```
#include<stdio.h>
#include<conio.h>
void main(){
    int a[10], n, i, low, high, mid, key, found=0;
    clrscr();
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter sorted elements:\n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("Enter element to search: ");
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<=high){
        mid=(low+high)/2;
        if(a[mid]==key){
            found=1;
            break;
        }
        else if(a[mid]<key){
            low=mid+1;
        }
        else{
            high=mid-1;
        }
    }
}
```

```

    }
}

if(found)

    printf("Element found at position %d",mid+1);

else

    printf("Element not found");

    getch();

}

```

Output:

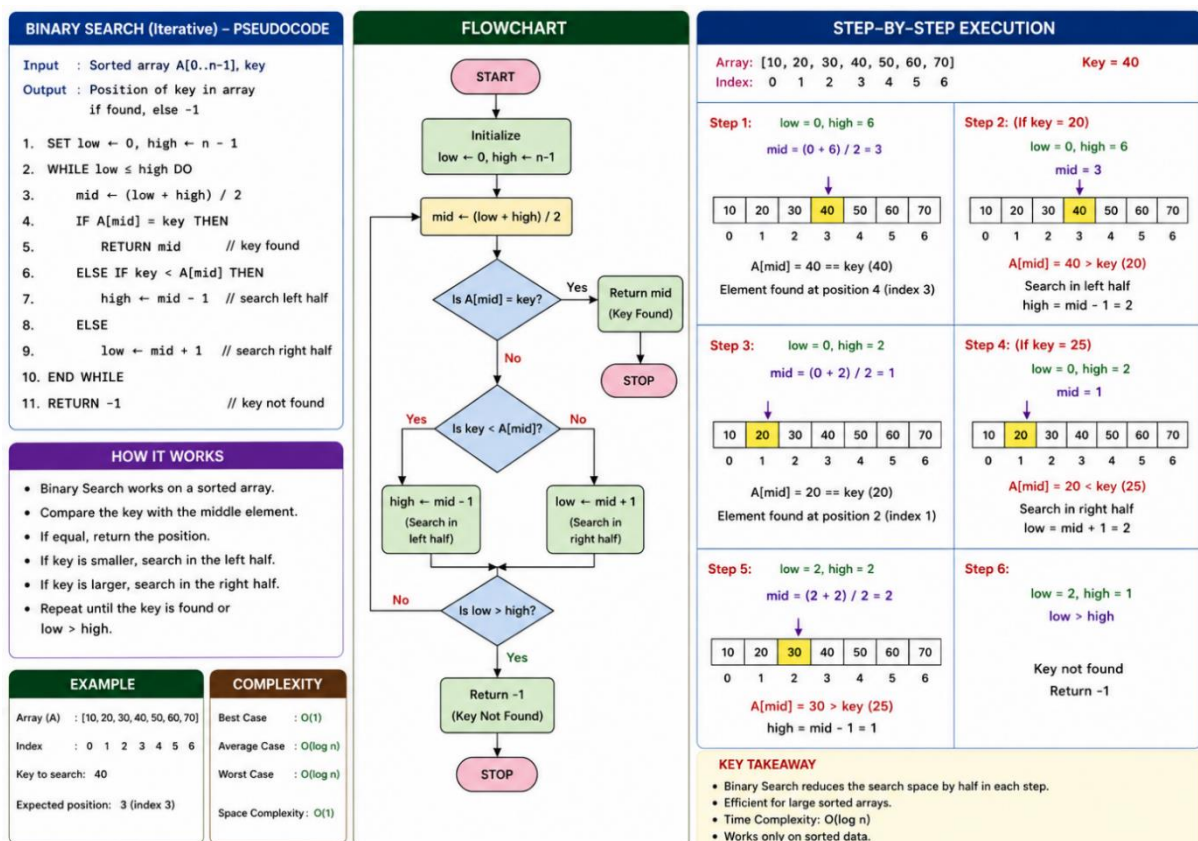
Enter number of elements: 5

Enter sorted elements:

10 20 30 40 50

Enter element to search: 30

Element found at position 3



2. Program to Implement Linear Search Using C

```
#include<stdio.h>

#include<conio.h>

void main(){

    int a[100], n, i, key, found = 0;

    clrscr();

    printf("Enter number of elements: ");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i=0; i<n; i++){

        scanf("%d", &a[i]);

    }

    printf("Enter element to search: ");

    scanf("%d", &key);

    for(i=0; i<n; i++){

        if(a[i] == key){

            found = 1;

            break;

        }

    }

    if(found)

        printf("Element found at position %d", i+1);

    else

        printf("Element not found");

    getch();

}
```

Output: Enter number of elements: 5

Enter elements:

10 25 30 45 50

Enter element to search: 45

Element found at position 4

2. LINEAR SEARCH

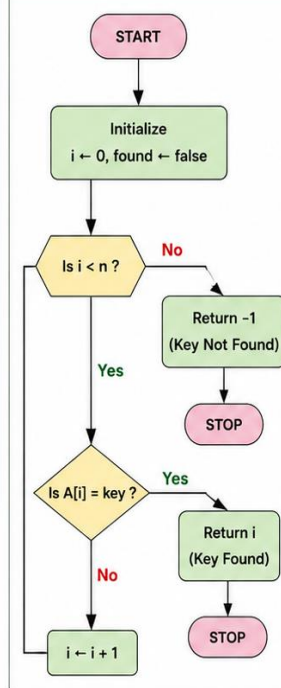
PSEUDOCODE

Input : Array A[0..n-1], key

Output : Position of key in array
if found, else -1

```
1. SET found ← false
2. FOR i ← 0 TO n-1 DO
3.   IF A[i] = key THEN
4.     RETURN i    // key found
5.   END IF
6. END FOR
7. RETURN -1      // key not found
```

FLOWCHART



SAMPLE OUTPUT

Enter number of elements: 5
Enter elements:
10 25 30 45 50
Enter element to search: 45
Element found at position 4

HOW IT WORKS (DRY RUN)

Array: [10, 25, 30, 45, 50]

Key = 45

Index: 0 1 2 3 4

Step 1: i = 0

10	25	30	45	50
0	1	2	3	4

Compare A[0] = 10 with key (45)

10 ≠ 45

Not found. Move to next element.

Step 2: i = 1

10	25	30	45	50
0	1	2	3	4

Compare A[1] = 25 with key (45)

25 ≠ 45

Not found. Move to next element.

Step 3: i = 2

10	25	30	45	50
0	1	2	3	4

Compare A[2] = 30 with key (45)

30 ≠ 45

Not found. Move to next element.

Step 4: i = 3

10	20	30	45	50
0	1	2	3	4

Compare A[3] = 45 with key (45)

45 = 45

Key found at index 3.

Step 5: Search stops and returns index 3.
(Position = 3 (index 3))